

*matter - anything that has mass and takes up space

-mass - the amount of matter in an object

-volume - the amount of space an object takes up

*density - the mass per volume of an object ($d=m/v$)

*atom - smallest part of matter, building block of matter

-nucleus - center of an atom, contains all the mass of the atom/atomic mass

-protons - positive charged particles, has mass, gives the atomic number

-neutron - neutral particle/no charge, has mass

-electron cloud, rings, shells, energy levels

-electron -negative charged particle, has no mass

*element - matter made of only one type of atom

O = oxygen

C = carbon

He = helium

H = hydrogen

N = nitrogen

Fe = iron

Na = sodium

Cl = chlorine

Ca = calcium

Au = gold

Ag = silver

Al = aluminum

*compound - two or more elements joined together to make a new substance

- compounds are different than the elements they're made of

-molecule - smallest part of a compound

H₂O = water

CO₂ = carbon dioxide

NaCl = sodium chloride/salt

C₆H₁₂O₆ = glucose(sugar)

*periodic table - an organized chart of all elements

-groups/families are the vertical columns, they're organized by similar properties or characteristics of elements

-periods are the horizontal rows, they're organized by increasing atomic number

-atomic number = the number of protons

-atomic mass = the number of protons and neutrons

-every elements symbol and name

*solid - has definite shape and volume, particles are close together and moving slow (vibrating)

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*liquid - has definite volume, takes the shape of container, particles far apart and moving fast

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*gas - has no definite shape or volume, particles are very far apart and moving very fast

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